Title of the paper: Can Deep Trade Agreements Absorb Geopolitical Shocks? Global Evidence from Intermediate-Goods Supply Chains

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ABSTRACT

Does institutional depth in preferential trade agreements insulate cross-border production networks against geopolitical friction? Analyzing a global panel of intermediate-goods trade from 2007 to 2023, this study investigates how agreement depth conditions supply-chain resilience under geopolitical risk. Using a structural gravity framework alongside duration models of trade-link survival, we uncover a sharp divergence across margins that challenges the conventional buffering view. First, contrary to the premise of institutional protection, trade between preferential agreement partners contracts more severely than trade between non-partners when global geopolitical risk rises. Second, at the extensive margin, deeper agreements accelerate rather than prevent relationship termination. Rather than an indiscriminate collapse, this attrition reflects selective pruning: elevated uncertainty leads suppliers to shed small-scale, marginal linkages burdened by fixed administrative compliance and restrictive rules-of-origin requirements. Third, institutional depth provides meaningful buffering only when agreements geographically align with bilateral, pair-specific geopolitical exposure, but fails to mitigate common global shocks. These disruptions are concentrated in contract-intensive manufacturing but leave post-exit recovery horizons intact. The findings demonstrate that deep integration does not inherently confer supply-chain resilience, offering critical caveats for contemporary "friend-shoring" policies.

Keywords: Deep trade agreements; Geopolitical risk; Intermediate goods; Global value chains; Trade-link survival; Gravity model

1. Introduction

Geopolitical risk has become a persistent source of instability for international production networks. Successive episodes since 2014, from the sanctions on Russia to the war in Ukraine, have made tensions between states a recurrent feature of the trading environment (Caldara & Iacoviello, 2022), and trade is reorganising along bloc lines in response (Campos et al., 2023; Gopinath et al., 2025). The consequences reach well beyond the policy measures themselves: in the 2014 sanctions episode the sanctioning countries lost approximately US$42

billion in exports, most of it in products no retaliatory measure had targeted (Crozet & Hinz, 2020). For firms embedded in cross-border production, such shocks mean cancelled orders of foreign inputs and, at the limit, the end of supplier relationships (Boehm et al., 2019; Novy & Taylor, 2020).

Trading relationships do not respond uniformly to such shocks, however. Transmission is state-dependent, shaped by the relationship-specificity of the inputs exchanged (Nunn, 2007; Rauch, 1999), the short-run substitutability of suppliers (Barrot & Sauvagnat, 2016) and the concentration of the trade network (Kramarz et al., 2020). In this context, the institutional depth of the preferential trade agreement governing a relationship emerges as an important but still unexamined condition. Agreements have grown steadily in legally enforceable content (Hofmann et al., 2019), and the case for "friend-shoring" rests on the premise that institutional commitment makes the supply chains between partners more resilient (Javorcik et al., 2024). Importantly, because global geopolitical risk is a common shock that reaches every country pair in the same year, its effect on a relationship depends less on the level of risk than on how the pair is structurally exposed to it. Identification therefore requires a cross-sectional rather than a purely time-series approach.

Institutional depth may shape this process in either direction. Viewed as endogenously incomplete contracts (Horn et al., 2010), deeper agreements close gaps in commitment and remove uncertainty about future policy (Handley & Limão, 2017), and so should dampen the response of trade to a shock. Viewed through the global-sourcing model (Antràs & Helpman, 2004), the same provisions induce the relationship-specific investment that constitutes exposure, and the fixed cost of maintaining preferential status (Anson et al., 2005) falls hardest on the smallest relationships once uncertainty lowers their expected value. Depth may therefore matter not as an isolated shock absorber but as a structural condition whose sign cannot be assumed in advance (Grossman et al., 2023).

Despite the growing literature on geopolitical risk and on deep trade agreements, three gaps remain. First, estimates of the trade effects of geopolitical risk are obtained for countries or broad sectors (Bloom, 2009; Caldara & Iacoviello, 2022), whereas the mechanisms they invoke operate between a particular pair trading a particular product. Second, the deep-agreement literature estimates level effects in ordinary times (Laget et al., 2020; Mattoo et al., 2022; Osnago et al., 2019) and has not asked whether depth conditions the response of trade to a shock. Third, most evidence concerns trade values, although the survival of trade relationships is where fixed costs bite (Besedeš & Prusa, 2006a; Melitz, 2003), and the two margins may respond with opposite signs. To address these limitations, this study adopts a heterogeneity-based identification strategy on a panel of bilateral intermediate-goods trade covering 160 economies, 88 HS two-digit chapters and 2007–2023 (16.75 million observations), combined with the count of legally enforceable provisions in the World Bank Deep Trade Agreements database and the Caldara and Iacoviello (2022) index. Because global risk is absorbed by time-varying fixed effects, the objective is not the average effect of risk over time. Instead, the analysis exploits cross-sectional variation in agreement membership and depth, at the trade-value margin through a Poisson gravity model and at the extensive margin through linear probability models of link survival and post-exit recovery.

Within this framework, global geopolitical risk serves as a contextual variable rather than an independently identified treatment. The central question is not whether geopolitical risk reduces intermediate-goods trade, but why some trading relationships contract, or disappear, more than others when exposed to the same shock. The study addresses four research questions:

1. Does intermediate-goods trade between preferential-agreement partners contract by more than trade between comparable non-partner pairs when global geopolitical risk rises, and does the differential follow the sectoral ordering that relationship-specificity predicts?

2. Does agreement depth dampen or amplify the response of trade values to geopolitical risk?

3. Does the conditioning effect of depth differ between the trade-value margin of trade values and the extensive margin of trade-link survival?

4. Is any adverse effect of depth on post-exit re-entry transitory or permanent?

This study contributes to the literature in three ways. First, it provides global pair–product evidence on the interaction between geopolitical risk and agreement depth on both margins of trade. Trade between preferential partners falls by a further 2.38 percent per standard deviation of global risk, concentrated in differentiated manufacturing, while depth has no distinguishable average effect on trade values. Second, it carries a common-shock identification strategy to the extensive margin, where each additional ten enforceable provisions lower one-year link survival by 0.52 percentage points per standard deviation of risk, the most stable estimate in the paper. Third, it shows that resilience is margin-dependent: under country-specific risk, depth becomes protective for trade values while remaining adverse for link survival, a selective pruning of marginal relationships driven by fixed compliance costs and transitory in its effect on re-entry. An evaluation of deep agreements that measures only trade value will therefore record insurance while the network beneath it thins.

The remainder of the paper proceeds as follows. Section 2 develops the hypotheses, Section 3 describes the data and empirical strategy, Section 4 reports the results, and Section 5 concludes.

2. Literature Review and Hypothesis Development

2.1. Geopolitical Risk and Supply-Chain Interdependence

Geopolitical risk (GPR) – the risk associated with wars, terrorist acts, and inter-state tensions, measured by newspaper coverage of adverse events (Caldara & Iacoviello, 2022) – differs from economic policy uncertainty (Baker et al., 2016) in originating between states, beyond any single government's control. Real Options Theory (Dixit, 1989) supplies the channel: with sunk costs, uncertainty confers value on waiting, so second-moment shocks alone depress investment and hiring (Bloom, 2009; Caldara & Iacoviello, 2022).

Intermediate goods are more sensitive than final goods for three reasons: firms holding inventories of imported inputs cut foreign orders disproportionately under uncertainty (Novy & Taylor, 2020); input relationships rest on non-contractible, relationship-specific investments whose quasi-rents are destroyed when a link is severed (Antràs, 2003; Antràs & Helpman, 2004; Nunn, 2007); and inputs are hard to substitute in the short run, so disruptions pass nearly one-for-one to output and downstream customers (Boehm et al., 2019; Barrot & Sauvagnat, 2016). Preferential integration deepens each of these features – rules of origin redirect sourcing toward members (Conconi et al., 2018) – removing outside options as it deepens commitment, and so generates exposure rather than protection. If relationship-specificity is the channel, vulnerability should vary across products: homogeneous goods traded on organised exchanges can be re-sourced at market prices, whereas differentiated goods depend on established networks (Rauch, 1999).

This remains untested because existing estimates are aggregate, and because global GPR, common to all countries in a year, is absorbed by time-varying fixed effects; identification must therefore come from cross-sectional exposure, here preferential integration, interacted with the shock.

Hypothesis H1 (Supply-Chain Exposure): Geopolitical risk contracts intermediate-goods trade more among preferential trade partners than non-partners, especially in GVC-complex industries.

2.2. The Conditioning Role of Deep Trade Agreements

Trade agreements have deepened steadily, with growth concentrated in legally enforceable provisions on competition, investment, and intellectual property (Hofmann et al., 2019). Whether depth changes how trade responds to shocks is a question on which two literatures give opposite answers.

The buffering view treats agreements as endogenously incomplete contracts (Horn et al., 2010): each enforceable provision closes a gap left by costly contracting, so depth remedies

vulnerability. Binding commitments raise trade and entry even without tariff changes because they remove uncertainty about future protection (Handley & Limão, 2017); dispute settlement gives firms recourse when a partner's conduct changes – precisely the contingency a geopolitical shock creates; and deeper agreements raise value-chain trade and vertical FDI (Laget et al., 2020; Orefice & Rocha, 2014; Osnago et al., 2019; Mattoo et al., 2022). Read forward, depth should dampen sensitivity to GPR.

The exposure view reads the same evidence differently: depth causes deeper, more relationship-specific integration – the depth–vertical-FDI link runs through provisions that improve the contractibility of supplier inputs (Osnago et al., 2019), the Global Sourcing mechanism of Antràs and Helpman (2004) at work – and relationship-specific investment is both the source of gains from integration and the definition of vulnerability to disruption. Two mechanisms reinforce this. Using preferential treatment may involve fixed administrative costs that are large relative to small shipments. For NAFTA, Anson et al. (2005) estimate average compliance costs of about 6 percent ad valorem against an average preference margin of 4 percent. Evidence from the US Generalized System of Preferences and the US–Colombia FTA also shows that preference utilisation increases with shipment size and duty savings, consistent with fixed utilisation costs (Hakobyan, 2015; Benguria, 2022). Trade networks may also be more concentrated, and therefore more exposed to disruptions, than diversification arguments assume (Kramarz et al., 2020; Elliott et al., 2022). Caselli et al. (2020) show that openness can reduce volatility by diversifying country-specific shocks, but this benefit may be weaker for common global shocks that affect many partners simultaneously.

Because the two views differ only in conditional responses, and theory offers no unambiguous sign for resilience policy (Grossman et al., 2023), the hypothesis treats the sign as the object of estimation.

Hypothesis 2a (Institutional Buffering): Greater trade agreement depth mitigates the adverse trade impact of geopolitical risk by reducing policy uncertainty and providing binding dispute settlement.

Hypothesis 2b (Institutional Inflexibility): Greater trade agreement depth amplifies the adverse trade impact of geopolitical risk by entrenching relationship-specificity and imposing rigid compliance costs.

2.3. Extensive Margin and Trade-Link Survival

Trade relationships are far more fragile than aggregate flows suggest: the median export spell lasts two to four years, with strong negative duration dependence (Besedeš & Prusa, 2006a; Nitsch, 2009; Brenton et al., 2010), and differentiated products sustain longer relationships than homogeneous ones (Besedeš & Prusa, 2006b; Obashi, 2010). Because the hazard concentrates among young and small relationships, any force that raises the survival threshold bites there first.

The two margins can diverge under a common shock because fixed costs govern whether a relationship is maintained, not how much is sold through it: they act through a participation cut-off (Melitz, 2003), widened by uncertainty into a band of inaction (Dixit, 1989) and made binding by large sunk costs (Das et al., 2007). A shock may thus cut the extensive margin while leaving the trade-value margin undisturbed. Combined with fixed compliance costs, this yields selective pruning: because maintaining preferential status costs the same regardless of relationship size, relationships near the threshold fail first when uncertainty lowers expected returns. Deeper agreements raise that cost, and may therefore protect the value of large relationships while eliminating marginal ones. Relaxing rules of origin, beneficial in aggregate, leads the least productive exporters to exit (Bombarda & Gamberoni, 2013), and firms abandon a preference altogether beyond a threshold of origin stringency (Head et al., 2024).

With sunk costs, exiting firms do not return and a temporary shock changes market structure permanently (Baldwin & Krugman, 1989); under transitory friction re-entry resumes

once the shock passes, so a one-year re-entry effect that disappears by two years indicates friction rather than hysteresis. Policy shocks raise the hazard of relationship death (Besedeš & Prusa, 2013); whether depth conditions survival under risk shocks, and whether the two margins respond with opposite signs, has not been tested.

Hypothesis 3 (Two-Margin Divergence & Selective Pruning): Under geopolitical risk, agreement depth reduces trade-link survival at the extensive margin while conditional mean trade values are preserved at the trade-value margin.

3. Methodology

3.1. Data and sample construction

Trade data are CEPII BACI (release V202601), six-digit HS 2007 aggregated to HS two-digit chapters over 2007–2023, which reconciles mirrored exporter and importer declarations into one harmonised flow. Agreement membership and depth are from the World Bank Deep Trade Agreements (DTA) database, with identifiers reconciled against the WTO Regional Trade Agreements database because content is merged onto membership by RTA identifier rather than by name. Geopolitical risk is the Caldara and Iacoviello (2022) index.

The sample comprises the 160 largest economies active as both exporters and importers in every year, covering 99.83% of global BACI exports and yielding 12,720 unique bilateral pairs, or 432,480 directed pair-year cells. Flows are classified by end use via the UN Broad Economic Categories (BEC Rev. 4) correspondence. Intermediate goods, the analytical focus, comprise industrial supplies, food and beverages for industry, primary and processed fuels other than motor spirit, and parts and accessories of capital and transport equipment. Because misclassified end use would contaminate the product-level outcome itself, a non-discretionary rule leaves unclassified the BEC categories spanning more than one end use, the ambiguously mapped lines, and one unmapped petroleum line carrying 4.11% of trade value; the filter discards roughly 9.5% of total trade value. Intermediate goods are 57.19% of retained value, consumption and capital goods 17.14% and 16.21%.

Three panels underlie the analysis (Table 2). The intensive-margin panel tracks trade value across 985,562 pair–chapter groups in 88 chapters with positive trade in at least one year, retaining every zero-trade year within them (47.52% zeros). The extensive-margin panel evaluates link exit conditional on positive trade in t−1 (unconditional survival rate 84.94%). The post-exit recovery panel follows the 1,237,520 exits from that risk set over one- and two-year windows. A fourth, restricted subsample covers the 44 economies with national geopolitical risk series – 946 pairs, 2,520,781 trade-value observations and 1,885,368 link-years; because these economies are systematically larger and better covered, the global-risk benchmark is re-estimated inside it rather than compared across samples.

Estimation samples are marginally smaller than these panels because fixed-effect separation and singleton removal discard observations with no identifying information – 105 rows at the trade-value margin and 174,448 in the survival panel (Correia et al., 2020). Pair–chapter groups with zero trade in all seventeen years are perfectly separated by their fixed effects; 266 of the 12,720 unordered pairs are thereby excluded, leaving 12,454 pair clusters, so intensive-margin parameters are conditional on a relationship having existed during the sample period and do not govern the entry of new linkages.

Table 1. Variable definitions and data sources

Variable	Definition	Source
TradeValue	Bilateral intermediate-goods exports from economy i to j, aggregated to HS2, in millions of current U.S. dollars	CEPII BACI HS2007 V202601; UN BEC Rev. 4 correspondence
Survival	One when a pair–HS2 relationship with positive trade in t−1 remains positive in t	CEPII BACI

Variable	Definition	Source
Reentry1Y, Reentry2Y	One when an exited link records positive trade again within one (two) years	CEPII BACI
FullRecovery1Y, FullRecovery2Y	One when post-exit trade returns to at least its pre-exit value within one (two) years	CEPII BACI
GlobalGPR	Annual mean of the global Geopolitical Risk Index, standardised on the 1985–2019 base period of the published index	Caldara and Iacoviello (2022)
GPRThreat, GPRAct	Standardised annual means of the threat and acts components	Caldara and Iacoviello (2022)
GPRPeak, LogGPR	Standardised within-year maximum, and standardised log of one plus the annual mean	Caldara and Iacoviello (2022)
PairGPRAvg, PairGPRMax, PairGPRLogAvg	Average, maximum and log-average of standardised exporter- and importer-specific risk indices per pair-year	Country-specific GPR series
FTA	One when at least one preferential agreement was in force for the pair in the preceding year	World Bank DTA Database, WTO RTA Database
DTAdepth	Legally enforceable policy areas in active agreements in the preceding year, divided by ten; overlapping agreements combined by the provision-level maximum	World Bank DTA Horizontal Content Database

Variable	Definition	Source
BundleShare	Share of a policy bundle's legally enforceable provisions in force in the preceding year, in [0,1]; five bundles, defined in Section 3.2	World Bank DTA Horizontal Content Database
GPR × FTA, GPR × DTAdepth	Interactions of the risk shock with lagged membership and depth	Authors' calculations
LaggedTrade, PreExitTrade	Log of one plus trade value in the preceding year, and in the year preceding exit	CEPII BACI

Source: Authors' compilation

3.2. Variable measurement

The intensive-margin outcome is TradeValue, denoted X_{ijkt} : intermediate-goods exports from economy i to economy j in HS2 chapter k and year t, in millions of current US dollars. These four indices carry the same meaning throughout. The outcome enters in levels with zeros retained, subject to the group-level support restriction of Section 3.1.

The extensive-margin outcomes are indicators on the same flow. A link is at risk in year t when it traded in t-1, and survives if it still trades in t:

$$Survive_{ijkt} = 1[X_{ijkt} > 0] \; conditional \; on \; X_{ijkt-1} > 0$$

For links that exit, two further outcomes are evaluated over horizons $\tau \in \{1,2\}$ years:

$$Recenter_{ijkt}^{T} = 1\left[\max_{s=1,\ldots,T} X_{ijk,t+s} > 0\right], \quad FullRecovery_{ijkt}^{T} = 1\left[\max_{s=1,\ldots,T} X_{ijk,t+s} \geq X_{ijk,t-1}\right]$$

The two outcomes are nested: a relationship can return without recovering its former size, but cannot recover without returning, so the gap between them measures the depth of the disruption. Table 2 reports the unconditional rate of each. Measuring relationship outcomes on

discrete annual spells follows the trade-duration literature (Besedeš & Prusa, 2006a; Nitsch, 2009).

The key explanatory variable is global geopolitical risk, GlobalGPR in Table 1 and GPR_t below. The index of Caldara and Iacoviello (2022) measures the share of newspaper articles discussing adverse geopolitical events, and is published monthly alongside a threat and an act component. Monthly values are aggregated to the year and standardised:

$$GPR_t = \frac{G\bar{P}R_t - \mu_{ref}}{\sigma_{ref}}, \quad G\bar{P}R_t = \frac{1}{12}\sum_{m=1}^{12} GPR_{mt}, \quad \mu_{ref} = 100, \quad \sigma_{ref} = 28.01$$

with the mean and standard deviation of the annual series over 1985–2019. That window is not an analyst's choice. Caldara and Iacoviello construct the index to average exactly 100 over it, and the same base carries through to both components. Standardising on it therefore re-expresses the published index on its own scale, and avoids rescaling the shock by the very turbulence the study examines.

Geopolitical risk is not interchangeable with economic policy uncertainty in the sense of Baker et al. (2016), although the two are often used as substitutes. Policy uncertainty concerns domestic fiscal, monetary and regulatory decisions, which a government can in principle reduce through its own commitments; geopolitical risk arises between states, and no single government can resolve it unilaterally. That asymmetry is what makes bilateral institutions a candidate remedy, and therefore what makes the conditioning test meaningful.

Four alternative measures enter the robustness programme of Section 3.3.3: the threat and act components (GPRThreat, GPRAct), the within-year maximum (GPRPeak), and the log of one plus the annual mean (LogGPR). Because global risk is common to all pairs within a year, a cross-sectional counterpart is also built from 44 national series. Letting g_{it} denote the standardised national index of economy i,

$$PairGPRAvg_{ijt} = \tilde{z}\left(\frac{1}{2}(g_{it} + g_{jt})\right), \quad PairGPRMax_{ijt} = \tilde{z}(\max\{g_{it}, g_{jt}\})$$

where $\tilde{z}(\cdot)$ re-standardises across covered pair-years; a log-average variant, PairGPRLogAvg, is built the same way from the logged national indices. The average captures joint exposure, the maximum exposure through the more troubled partner. They correlate at 0.97 with one another but only 0.62 and 0.59 with the global index, which makes them an informative alternative rather than a restatement of the same time-series variation. The two scales are not interchangeable. The global index, standardised on its 1985–2019 history, has a sample standard deviation of 0.68 over 2007–2023; the pair measures have one of exactly 1 by construction. A one-unit movement in a pair measure is therefore roughly 1.5 times a one-unit movement in the global measure in within-sample terms, and coefficients on the two shocks are compared with that ratio in view.

The two moderating variables are agreement membership and agreement depth. FTA equals one from the calendar year in which a preferential agreement enters into force for the pair, as recorded in the World Bank bilateral-membership table, and remains one while it is in force. DTAdepth measures how much of a pair's policy space is bound by enforceable commitment. The World Bank horizontal-content database codes 52 policy areas twice for every agreement: once for whether the agreement covers the area, once for whether the commitment is legally enforceable (Hofmann et al., 2019). Writing $A_{ij,t-1}$ for the agreements in force between i and j, and $LE_{pa} \in \{0,1\}$ for whether area p is legally enforceable under agreement a,

$$Depth_{ij,t-1} = \frac{1}{10}\sum_{p=1}^{52} \max_{a \in A_{ij,t-1}} LE_{pa}$$

Three features shape what the coefficient means. First, the summand counts only enforceable areas. This follows from the mechanism of Section 2.2: buffering operates through binding commitment and dispute settlement, both of which presuppose enforceability. Counting areas merely mentioned in an agreement's text would credit buffering capacity to language creating no obligation, and would tie measured depth to agreement verbosity. Provision

counting of this kind is the standard operationalisation in work using this database (Hofmann et al., 2019; Osnago et al., 2019; Laget et al., 2020; Mattoo et al., 2022). It is preferred to DESTA (Dür et al., 2014), whose depth measures are built from the presence of provisions rather than from a per-area enforceability code. DESTA does code dispute settlement in detail, but as a separate agreement-level dimension rather than as an attribute of each policy area, and it is that attribute the mechanism here requires. Second, overlapping agreements are combined by the maximum over a rather than by summation, because a pair belonging to both a regional and a bilateral agreement covering the same area is bound on it once, not twice. The bilateral table is collapsed to unordered pairs beforehand, since content is symmetric. Third, the count is divided by ten, so that a unit coefficient reads as ten additional enforceable areas; the effect of a single provision would be too small to report legibly.

Among active agreements the distribution is right-skewed, median 11 enforceable areas, mean 12.68, maximum 38 of the 52 coded. The scaled variable therefore ranges from 0 to 3.8, with an unconditional mean of 0.53, since it is zero for pairs without an agreement. For the mechanism analysis the same indicators are aggregated into five bundles b, specified in advance by the authors, grouping the policy areas as they are defined in the World Bank handbook (Mattoo et al., 2020) rather than reproducing a taxonomy given there:

$$BundleShare_{b,ij,t-1} = \frac{1}{|b|} \sum_{p \in b} \max_{a \in A_{ij,t-1}} LE_{pa} \in [0,1]$$

so a coefficient describes movement from none to all of a bundle's provisions. The bundles are border coordination; investment and production; digital and knowledge; services and procurement; and environmental provisions. The last contains a single provision, so its bundle-level and provision-level estimates are the same quantity at two levels of aggregation, not two independent findings. All agreement variables are measured at $t-1$, with 2006 as the lag base for the 2007 outcome, so membership and depth are predetermined relative to the flow they explain.

The survival and recovery models include one control, LaggedTrade, defined as lagged link size $ln(1 + X_{ijk,t-1})$, with the recovery models using the pre-exit value instead. The log of one plus the flow is used rather than the level because the size gradient in survival is strongly concave, and because the transformation remains defined at the boundary of the at-risk set. The control is necessary on two counts: small links fail more often than large ones for reasons unrelated to policy, and survival rises with accumulated exporting experience (Besedeš & Prusa, 2006a; Brenton et al., 2010). Without it the interaction terms would partly absorb composition rather than response. No further covariates are needed, since the fixed effects below absorb everything time-invariant at pair–chapter level and everything time-varying at exporter, importer and product level. Two measurement issues are handled by exclusion rather than imputation. Pair-years in which any agreement in force lacks content coding are excluded from the depth baseline, 26 directed pair-years at the lagged value, or 2,093 rows of the panel, and product lines that cannot be assigned unambiguously to a single end use are excluded rather than allocated by judgement. Both preserve the interpretation of the treatment variables at a small cost in coverage, the right trade-off when the variable in question is the object of interest rather than a control.

Table 2 reports descriptive statistics. The extreme right skew of trade value, with a median of zero, is what makes Poisson estimation in levels appropriate.

Table 2. Descriptive statistics

Variable	N	Mean	SD	Min	Median	Max
Intermediate-goods trade value (USD million)	16,752,461	10.303	276.895	0	0	154,728.17

Variable	N	Mean	SD	Min	Median	Max
FTA membership, $t-1$	16,752,461	0.358	0.479	0	0	1
Legally enforceable DTA depth, $t-1$	16,752,461	0.531	0.970	0	0	3.800
Global GPR, standardised	16,752,461	-0.132	0.683	-0.811	-0.323	2.056
Lagged link size, ln(1 + Trade $t-1$)	8,219,579	0.682	1.246	0	0.077	11.949
Trade-link survival	8,219,579	0.849	0.358	0	1	1
Post-exit re-entry within one year	1,152,080	0.341	0.474	0	0	1
Post-exit re-entry within two years	1,065,382	0.497	0.500	0	0	1
Full recovery within one year	1,152,080	0.171	0.376	0	0	1
Full recovery within two years	1,065,382	0.281	0.450	0	0	1

Source: Authors' compilation

3.3. Empirical strategy

3.3.1. Baseline specification: The trade-value margin

The intensive-margin model is a structural gravity equation estimated by Poisson pseudo-maximum likelihood:

$$X_{ijkt} = \exp\left[\beta_1\left(GPR_t \times FTA_{ij,t-1}\right) + \beta_2\left(GPR_t \times Depth_{ij,t-1}\right) + \beta_3 FTA_{ij,t-1}\right.$$
$$\left. + \beta_4 Depth_{ij,t-1} + \mu_{ijk} + \lambda_{it} + \theta_{jt} + \delta_{kt}\right] \times \varepsilon_{ijkt} \quad (1)$$

where μ_{ijk} are pair-chapter fixed effects, λ_{it} and θ_{jt} exporter-year and import-year effects, δ_{kt} chapter-year effects, and ε_{ijkt} a multiplicative error

Poisson estimation is appropriate on two grounds. Log-linearising a gravity equation yields inconsistent estimates under heteroskedasticity, since the expectation of a log differs from the log of an expectation, whereas the pseudo-maximum-likelihood estimator targets the conditional mean directly (Santos Silva & Tenreyro, 2006). It also accommodates zero flows without transformation, decisive given that 47.52% of observations are zeros.

The fixed effects serve distinct purposes. Pair–chapter effects absorb every time-invariant determinant of trade between a given pair in a given product category – distance, contiguity, shared language, colonial history, persistent bilateral complementarity. Exporter–year and importer–year effects absorb time-varying multilateral resistance and all national macroeconomic conditions, so the price indices required by theory are controlled non-parametrically rather than proxied (Anderson & van Wincoop, 2003; Head & Mayer, 2014; Yotov et al., 2016). Chapter–year effects absorb global product-specific shocks, including commodity price cycles that would otherwise be attributed to geopolitical risk. No standalone risk term is estimable, so the coefficients of interest are β_1 and β_2, describing how the trade of agreement-linked pairs moves relative to comparable non-agreement pairs as global risk rises.

Equation (1) provides the primary test of the first two of the three hypotheses in Section 2. A negative β_1 supports H1, that trade between agreement partners contracts by more than trade between comparable non-partners when risk rises. The coefficient β_2 tests H2, which is stated two-directionally, so either sign is a finding: positive supports the buffering view, negative the exposure view. The sign is estimated rather than imposed.

Three specifications are estimated: restricted versions of Equation (1) containing the FTA and the depth interaction respectively, each with its own main effect, and the joint

specification, in which the FTA coefficient measures the interaction at zero depth and the depth coefficient the incremental effect of content conditional on membership.

Every coefficient is estimated under two clustering schemes. Pair clustering allows for arbitrary correlation within a trading relationship over time. Adding a year dimension additionally allows for shocks common to all pairs in a year – the very kind this paper studies – but rests on only seventeen clusters in that dimension, where the asymptotics behind multiway clustering are weak (Cameron et al., 2011).

Neither scheme dominates the other on a priori grounds, and the two need not agree, so both are carried rather than one being designated correct. The tables report the two-way error throughout; where the schemes lead to different conclusions about a coefficient, Section 4 reports the divergence rather than selecting the more convenient of the two. Coefficients are converted to percentage effects as $100 \times (\exp(\beta) - 1)$.

3.3.2. The extensive margin: Trade-link survival

Equation (1) estimates the conditional mean of trade value in a panel that retains zero observations; it should therefore not be interpreted as a pure intensive-margin model conditional on positive trade. Equation (2) complements the trade-value specification by examining whether previously active links remain active or exit:

$$Survive_{ijkt} = \gamma_1\left(GPR_t \times FTA_{ij,t-1}\right) + \gamma_2\left(GPR_t \times Depth_{ij,t-1}\right) + \gamma_3 FTA_{ij,t-1} + \gamma_4 Depth_{ij,t-1} + \gamma_5 \ln\left(1 + X_{ijk,t-1}\right) + \mu_{ijk} + \lambda_{it} + \theta_{jt} + \delta_{kt} + u_{ijkt} \quad (2)$$

estimated on the at-risk sample. The fixed-effect structure is deliberately identical to Equation (1), because H3 is a claim about the difference between β_2 and γ_2, which would be uninterpretable if the two were conditioned differently. Five specifications are estimated: the FTA and depth interactions separately, each with and without the lagged-size control, and the joint specification with the control, which is the primary one – a depth effect that was merely a composition artefact would not survive that control.

Equation (2) tests part (a) of H3: a negative γ_2 alongside a non-negative β_2 would establish the predicted divergence between the margins. Sample composition provides the sharper test. If the mechanism is the selective pruning of Section 2.3, the effect must concentrate among small links, for which a size-invariant cost is proportionally largest, so restricting to established links should attenuate the coefficient.

Two alternatives are rejected, each for a different reason. A binary-response model with a non-linear link cannot be estimated consistently here, because the four sets of fixed effects enter the likelihood non-linearly and cannot be conditioned out. The resulting incidental-parameter problem would bias the interaction coefficients themselves, not merely their standard errors – and with 967,746 pair–chapter groups over at most sixteen years, the within-group ratio of parameters to observations is exactly the regime in which that bias is severe.

A proportional-hazards specification is rejected on two published grounds. Hess and Persson (2012) show continuous-time hazard models, and Cox in particular, are poorly suited to annual trade data, which are discrete and heavily tied. Brenton et al. (2010) show that substantial unobserved heterogeneity in product-level trade survival persists after extensive observable controls, and question the Cox-based estimates dominating the earlier literature.

A linear specification with pair–chapter fixed effects addresses both, absorbing the heterogeneity at issue while admitting a four-way structure a conditional hazard model could not accommodate at this scale. Its cost is that fitted probabilities are not bounded to the unit interval; because the parameters of interest are interactions evaluated near the sample mean this is unlikely to drive the results, but the limitation is acknowledged rather than assumed away.

$$\frac{\partial Survive_{ijkt}}{\partial GPR_t} = \gamma_1 FTA_{ij,t-1} + \gamma_2 Depth_{ij,t-1} \quad (3)$$

which is the reason the two coefficients must be read together. Both terms vanish for a pair with no agreement, so Equation (3) is evaluated at FTA=1, tracing $\gamma_1 + \gamma_2 Depth$ across the observed range of depth. Because γ_1 is the effect at zero depth, it is not the effect for a

typical agreement: the mean active agreement carries 12.68 enforceable provisions, so the depth term enters with substantial weight. The depth level at which the net effect crosses zero, and the level beyond which it is statistically distinguishable from zero, are read off this expression.

Part (b) of H3 concerns what happens after a relationship breaks, and requires following exits forward:

$$Y_{ijkt}^{T} = \pi_1\left(GPR_t \times FTA_{ij,t-1}\right) + \pi_2\left(GPR_t \times Depth_{ij,t-1}\right) + \pi_3 FTA_{ij,t-1} + \pi_4 Depth_{ij,t-1} + \pi_5 \ln\left(1 + X_{ijk,t-1}\right) + \nu_{ij} + \lambda_{it} + \theta_{jt} + \delta_{kt} + w_{ijkt} \quad (4)$$

where t indexes the exit year, Y_{ijkt}^{τ} is one of the four re-entry and recovery indicators evaluated over the τ years following that exit, $X_{ijk,t-1}$ is the pre-exit value, and λ_{it}, θ_{it}, δ_{it} are defined over the exit year; in the stringent specification ν_{ij} is replaced by pair–chapter effects ν_{ijk}. The full-recovery outcomes are defined relative to the same pre-exit value that enters as the size control, so π_5 absorbs the mechanical dependence of the recovery threshold on relationship size.

Equation (4) discriminates between the interpretations of Section 2.3. Under permanent hysteresis an adverse effect on re-entry should persist as the window lengthens; under transitory friction it should attenuate. The contrast between horizons is therefore the test, and neither coefficient is informative alone.

Each outcome is evaluated only on exits whose follow-up window falls inside the sample period, so the horizons rest on different event counts. Two fixed-effect specifications are reported, a baseline with directed-pair effects and a stringent one with pair–chapter effects. The stringent version is not simply a tighter baseline: it discards roughly a quarter of the events, because many pair–chapter cells contain too few exits to identify a separate effect, so differences between the two are read as a change in composition rather than as evidence about the parameter. Standard errors are clustered on the unordered pair and the exit year.

3.3.3. Mechanism identification and robustness checks

Equations (1) to (4) establish whether the conditioning effects exist and how they differ across margins. Three further exercises ask where they come from and whether they survive perturbation. The first re-estimates the baseline separately across ten pre-specified sectors defined by contiguous HS2 ranges, spanning agro-food inputs through precision manufactures:

$$X_{ijkt} = \exp\left[\beta_1^s\left(GPR_t \times FTA_{ij,t-1}\right) + \beta_2^s\left(GPR_t \times Depth_{ij,t-1}\right) + \beta_3^s FTA_{ij,t-1} + \beta_4^s Depth_{ij,t-1} + \mu_{ijk} + \lambda_{it} + \theta_{jt} + \delta_{kt}\right] \times \varepsilon_{ijkt}, \quad k \in s \quad (5)$$

This provides the evidence bearing on the scope clause of H1. If the transmission channel is relationship-specificity and low input substitutability, the ordering of β_1 should track asset specificity and value-chain complexity, largest in differentiated manufactures and smallest in homogeneous commodities. The exercise is a consistency test rather than a causal identification strategy: an ordering consistent with theory is difficult to reconcile with several rival explanations, but does not by itself establish the mechanism.

The second screens individual enforceable provisions and the five bundles of Section 3.2. Sixteen provision indicators are constructed, but two – information society and innovation policies – are in force for only 287 and 29 unordered pairs, too few to support a separate interaction, so the confirmatory screen covers the remaining fourteen. Bundles are estimated one at a time, standalone and conditional on FTA membership, never jointly, because inter-bundle correlations render joint estimates uninformative. The FTA-conditional form is the interpretable one, isolating agreement composition beyond mere membership.

Both exercises test many hypotheses simultaneously and are adjusted using the Benjamini–Hochberg procedure controlling the false discovery rate (Benjamini & Hochberg, 1995), at the 5% and 10% thresholds. The family is stated in each case, since an adjusted p-value has no meaning without it: the ten sectors, corrected separately for each interaction and each clustering scheme; the five bundles, corrected separately for the standalone and FTA-conditional specifications; and the fourteen provisions entering the confirmatory screen.

Two rules then govern how those adjusted p-values are read, both fixed in advance rather than chosen once the estimates are in hand. A sector surviving correction with a coefficient signed against the predicted ordering is reported rather than omitted, since it is evidence about the scope clause of H1 and not a nuisance. And provision- or bundle-level coefficients that reach conventional significance individually but fail correction are presented as exploratory; none of the paper's three hypotheses is permitted to rest upon them.

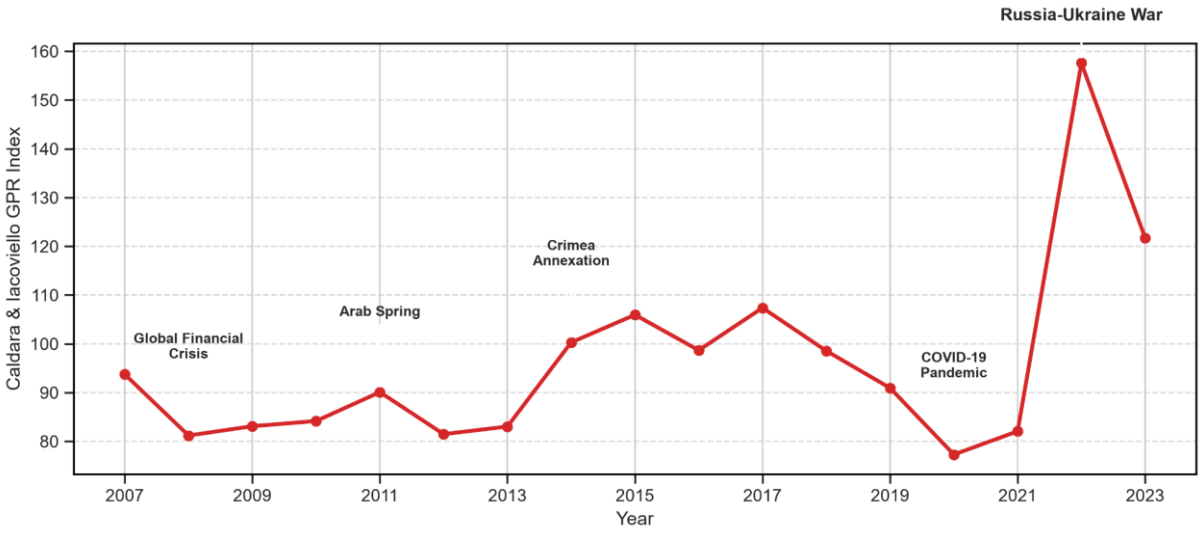
Four identification concerns warrant explicit statement. First, agreement depth is not randomly assigned. Countries choose whether and how deeply to integrate, and the relationship runs in both directions, since dense production-network trade itself raises the depth of subsequently negotiated agreements (Orefice & Rocha, 2014). Pair–chapter fixed effects absorb the time-invariant component of this selection, and lagging renders the treatment predetermined, but selection on time-varying unobservables cannot be ruled out and is carried into the limitations (Baier & Bergstrand, 2007). Second, Poisson estimation with several sets of fixed effects is subject to an incidental-parameter problem. For the three-way gravity model, Weidner and Zylkin (2021) show the estimator remains consistent for fixed T – and is the only estimator among a wide class of pseudo-maximum-likelihood gravity estimators that generally is – but that asymptotic confidence intervals are not correctly centred and cluster-robust variance estimates are biased. Their bias expressions are derived for the three-way case, while this specification adds a fourth dimension, so they are taken as indicative of the direction and rough magnitude of the problem rather than as an exact characterisation. Inference is treated cautiously in consequence. Third, the linear probability model carries the limitation discussed above. Fourth, the global shock is exogenous to any individual pair, which is the source of the design's credibility, but that exogeneity is bought precisely by giving up cross-sectional variation. Finally, the global shock is replaced by country-specific pair-level risk on the restricted subsample of Section 3.1. That subsample differs systematically from the full panel, so the comparison cannot be made against the full-sample baseline. The global-risk benchmark

is therefore re-estimated within the identical restricted sample and reported alongside it, and every statement contrasting the two measures is made against that internal benchmark, with the scale difference of Section 3.2 in view. Without this discipline, a change driven by sample composition could be misread as evidence about the shock measure.

4. Results and Discussion

This section presents the empirical findings in six stages, each building upon the previous one to construct a comprehensive narrative of how deep trade agreements and geopolitical risk jointly shape intermediate-goods trade flows. We begin with the baseline intensive-margin estimates, followed by sectoral heterogeneity analysis that reveals the economic mechanism of global value chain complexity. We then investigate the specific DTA policy provisions that serve as transmission channels, before examining the extensive margin through trade-link survival dynamics and post-exit recovery patterns. Finally, we confirm robustness by using country-specific geopolitical risk measures and synthesize the evidence into a unified narrative of mechanisms.

Figure 1. Global Geopolitical Risk (GPR) Index and Major Events (2007–2023)



Notes: Annual mean of standardized global GPR; key historical events are annotated.

Source: Authors' compilation based on CEPII BACI, World Bank DTA, and Caldara and Iacoviello (2022)

4.1. Baseline Results: The GPR–DTA Nexus on Intermediate-Goods Trade

Table 3 reports the baseline PPML gravity estimates for the interaction between standardized global geopolitical risk (GPR) and trade agreement variables on bilateral intermediate-goods exports at the HS2 product level. All specifications include the full set of pair–HS2 fixed effects, exporter–year fixed effects, importer–year fixed effects, and HS2–year fixed effects, thereby absorbing all time-invariant bilateral-product heterogeneity and all unilateral multilateral resistance terms consistent with the structural gravity literature (Anderson & van Wincoop, 2003; Head & Mayer, 2014). Standard errors are clustered at the unordered country-pair level, with multiway pair-plus-year clustering reported for inference robustness.

Column (1) estimates the interaction between GPR and a binary FTA indicator. The coefficient on GPR × FTA is −0.0241 ($p = 0.024$), implying that when global geopolitical risk rises by one historical standard deviation, intermediate-goods trade between FTA partners declines by an additional 2.38 percent relative to non-FTA pairs. This result is both statistically significant and economically meaningful: the magnitude is comparable to half the typical trade-creation effect of a shallow FTA reported in the meta-analysis of Head and Mayer (2014).

Column (2) replaces the binary FTA indicator with DTA depth, measured as the number of legally enforceable provisions divided by ten. The interaction GPR × DTA Depth yields a coefficient of −0.0053 ($p = 0.255$), which is negative but statistically insignificant. This suggests that, in isolation, the depth of institutional commitments does not systematically amplify geopolitical sensitivity at the aggregate level.

Column (3) enters both interactions jointly. The FTA interaction strengthens to −0.0382 ($p = 0.025$, effect: −3.75%), while the depth interaction turns positive at +0.0099 ($p = 0.202$, effect: +0.99%). Because this depth interaction is not statistically distinguishable from zero at conventional significance thresholds, we draw no confirmatory inference from its point

estimate. Furthermore, this sign reversal must be interpreted with econometric caution: the binary FTA indicator and the continuous depth measure exhibit an empirical correlation of 0.73 in our estimation panel, which constrains the precision with which their distinct interaction terms can be separately identified. We note the positive coefficient primarily because it conforms to the theoretical direction of an institutional buffering mechanism, and because an analogous trade-buffering effect becomes statistically significant when geopolitical risk is measured at the bilateral country-pair level (Table 8, Panel A) – an empirical pattern we examine systematically in Section 4.6.

Table 3. Baseline PPML Gravity Estimates: GPR × Trade Agreement Interactions

Variable	**(1) FTA interaction**	**(2) Depth interaction**	**(3) Joint FTA + depth**
GPR × DTA depth	–	−0.0053	0.0099
		(0.0047)	(0.0078)
		[−0.53%]	[+0.99%]
GPR × FTA	−0.0241**	–	−0.0382**
	(0.0107)		(0.0171)
	[−2.38%]		[−3.75%]
FTA level term	Included	–	Included
DTA-depth level term	–	Included	Included
Pair–HS2 fixed effects	Yes	Yes	Yes
Exporter–year fixed effects	Yes	Yes	Yes
Importer–year fixed effects	Yes	Yes	Yes
HS2–year fixed effects	Yes	Yes	Yes

Variable	(1) FTA interaction	(2) Depth interaction	(3) Joint FTA + depth
Observations	16,752,356	16,752,356	16,752,356

*Notes: PPML estimates with multiway-clustered standard errors in parentheses. Marginal effects in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$*

Source: Authors' compilation based on CEPII BACI, World Bank DTA, and Caldara and Iacoviello (2022)

4.2. Sectoral Heterogeneity: GVC Complexity as a Transmission Mechanism

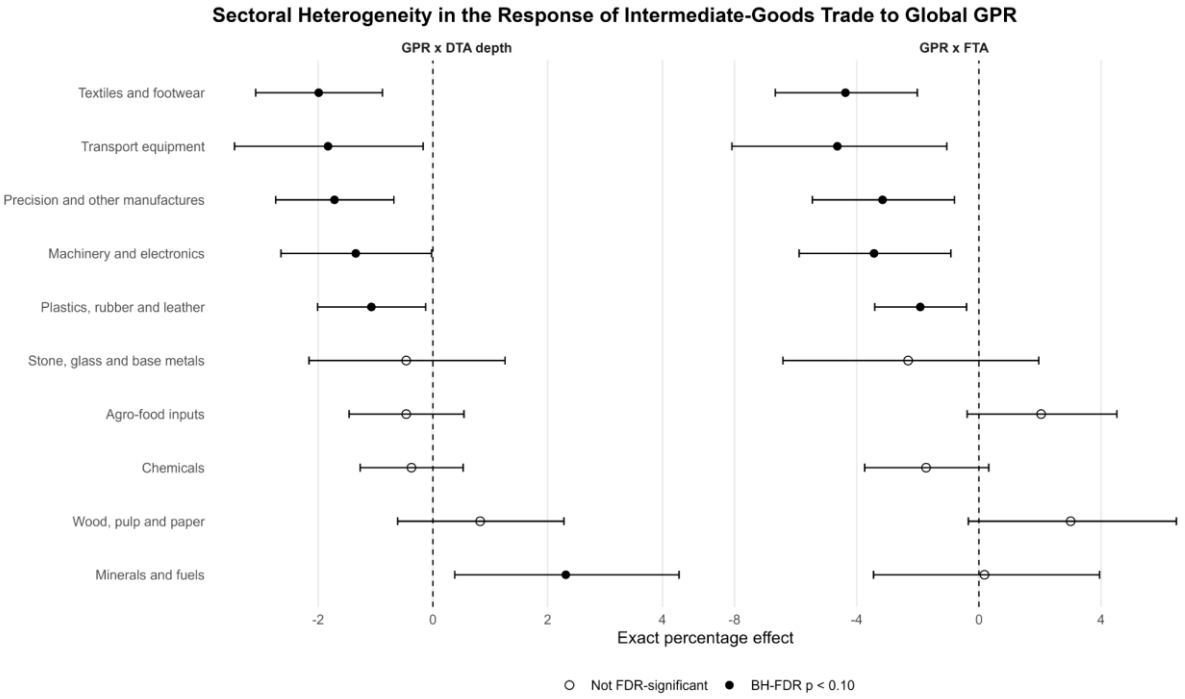
The aggregate baseline masks substantial heterogeneity across industries with different degrees of global value chain complexity, asset specificity, and input substitutability. To uncover the economic mechanism underlying the aggregate FTA–GPR interaction, we estimate sector-specific PPML models for each of ten pre-specified HS2 manufacturing groups. Table 4 and Figure 2 present the five core manufacturing sectors where both the GPR × FTA and GPR × DTA Depth interactions survive Benjamini–Hochberg false discovery rate (BH-FDR) correction at the 10 percent level. Appendix Table A1 reports the complete ten-sector estimation results across all manufacturing and primary commodity groups.

The sectoral pattern strongly supports the hypothesis that GVC complexity is the primary transmission channel through which geopolitical risk disrupts trade within preferential agreements. Transport equipment – an archetype of vertically fragmented production with extensive cross-border component sourcing – exhibits the largest FTA interaction (−4.63%, BH-FDR p = 0.026) and depth interaction (−1.83%, BH-FDR p = 0.062). Textiles and footwear, another sector characterized by long, geographically dispersed value chains and relationship-specific inputs, shows comparable magnitudes (FTA: −4.37%, BH-FDR p = 0.003; Depth: −1.99%, BH-FDR p = 0.005). Machinery and electronics (FTA: −3.43%; Depth: −1.34%) and precision instruments (FTA: −3.15%; Depth: −1.72%) follow a consistent pattern.

By contrast, sectors trading relatively homogeneous commodities with thick spot markets – agro-food inputs (+2.04%, BH-FDR p = 0.125), minerals and fuels (+0.18%, BH-FDR p = 0.923), and wood products (+3.00%, BH-FDR p = 0.125) – display positive or near-zero interactions that do not survive multiple-testing correction. These sectors feature low asset specificity, easy supplier switching, and active commodity exchanges, which insulate them from geopolitical disruptions within preferential frameworks.

This cross-sectoral gradient directly illuminates the transmission mechanism. The FTA–GPR sensitivity is not a generic feature of all preferential trade; rather, it is concentrated in industries where FTA-induced integration deepens Just-in-Time supply chains, raises relationship-specific investments, and creates hold-up vulnerabilities that geopolitical shocks exploit. This graded cross-sectoral pattern is consistent with the supply-chain interdependence channel and is difficult to reconcile with rival explanations that operate uniformly across all traded commodities, such as macroeconomic demand shocks or reverse causality in trade agreement formation. It should not, however, be construed as a definitive causal identification strategy: sectors characterized by higher GVC complexity systematically differ along other structural dimensions, including external financial dependence, cyclical durability, and geographic asset specificity, any of which could contribute to the observed cross-industry variation.

Figure 2. Sectoral Heterogeneity in the Response of Intermediate-Goods Trade to Global GPR



Notes: Point estimates and 95% confidence intervals. Filled circles indicate significance after Benjamini-Hochberg FDR correction at the 10% level

Source: Authors' compilation based on CEPII BACI, World Bank DTA, and Caldara and Iacoviello (2022)

4.3. Mechanism Analysis: DTA Policy Provisions as Micro-Level Shock Absorbers

Having documented substantial sectoral variation across production networks, we now investigate whether specific institutional provisions within deep trade agreements exhibit shock-absorbing properties, and if so, which substantive policy areas account for this capacity. This represents an exploratory inquiry: because the aggregate depth interaction in the baseline joint model (Table 3, Column 3) is not statistically distinguishable from zero, the analysis below constitutes a search for candidate micro-level mechanisms rather than a formal decomposition of an established aggregate effect.

4.3.1. DTA provision bundles

We classify sixteen legally enforceable provisions from the World Bank DTA database into five substantive thematic bundles: (i) Border Coordination (customs administration, sanitary and phytosanitary measures, and technical barriers to trade); (ii) Investment and

Production (investment disciplines, capital mobility, trade-related investment measures, and competition policy); (iii) Services and Procurement (trade in services and public procurement); (iv) Environmental Provisions (environmental laws and enforcement); and (v) Digital and Knowledge (data protection, intellectual property rights, research and technology cooperation, information society, innovation policies, and TRIPs-plus disciplines). Because inter-bundle correlations are substantial, we estimate separate PPML models for each individual bundle rather than entering all five jointly, thereby avoiding multi-collinear variance inflation (Appendix Table A2).

Among the five bundles, Environmental Provisions exhibit a nominally significant FTA-conditional interaction with GPR: a coefficient of +0.0384 (full-bundle effect: +3.91%, raw p = 0.049; Appendix Table A2). However, after applying Benjamini–Hochberg false discovery rate adjustment across the five bundle tests, the adjusted p-value rises to q = 0.244, failing to achieve statistical significance at conventional thresholds. We also note that the environmental bundle in our classification comprises a single enforceable provision (environmental laws); its point estimate and standard error are therefore identical to the individual provision result reported in Appendix Table A3. The Digital and Knowledge bundle displays a positive point estimate (+4.63%, raw p = 0.180, q = 0.451), while Border Coordination (+1.80%, q = 0.673), Investment and Production (+0.72%, q = 0.753), and Services and Procurement (+0.59%, q = 0.753) do not exhibit detectable shock-buffering properties.

4.3.2. Individual provision screening

To achieve greater institutional granularity, we screen fourteen of the sixteen coded provisions individually in FTA-conditional specifications (Appendix Table A3). Two provisions—information society disciplines (287 active pairs) and innovation policy provisions (29 active pairs)—are excluded prior to estimation because their empirical support falls below the pre-specified threshold of 500 unordered country pairs required for reliable cluster-robust estimation. Three provisions emerge with nominally significant positive interactions:

First, Competition Policy (+3.60%, p = 0.037) contributes to trade resilience by ensuring a level playing field that prevents dominant suppliers from exploiting crisis-driven market power. When geopolitical shocks create supply bottlenecks, enforceable competition disciplines discourage monopolistic price gouging and facilitate the entry of alternative suppliers, thereby maintaining trade flow continuity.

Second, Environmental Laws (+3.91%, p = 0.049) promote supply-chain transparency, traceability standards, and clean-production certification. These regulatory frameworks create shared compliance infrastructure that raises the costs of abrupt supply-chain reconfiguration, paradoxically anchoring trade relationships more firmly during geopolitical turbulence. Firms that have invested in meeting joint environmental standards face higher sunk costs of switching partners, reinforcing bilateral trade resilience.

Third, Research and Technology cooperation (+3.49%, p = 0.041) enhances firms' adaptive capacity by facilitating knowledge transfer, joint R&D, and technology licensing across borders. When geopolitical disruptions threaten input availability, firms embedded in technology-sharing frameworks can more rapidly develop substitute inputs, re-engineer production processes, or identify alternative sourcing routes – all of which sustain bilateral trade volumes.

Although competition policy (raw p = 0.037), environmental standards (raw p = 0.049), and research and technology cooperation (raw p = 0.041) achieve nominal significance at the 5 percent level, none of these fourteen individual provisions survives multiple-testing adjustment under the Benjamini–Hochberg procedure (q = 0.228 across all three candidate provisions; Appendix Table A3). In a testing family of fourteen simultaneous regressions, observing three nominal rejections at the 5 percent level is roughly consistent with the false discovery expectation under the joint null hypothesis. Consequently, we treat these specific institutional chapters as exploratory hypotheses for future investigation rather than established structural mechanisms. We highlight, nonetheless, an intriguing institutional pattern: all three candidate

disciplines represent 'behind-the-border' regulatory frameworks that address market contestability and adaptive capability, whereas none of the classical border disciplines (customs administration, SPS, TBT) or formal investment chapters exhibits any mitigating capacity.

4.4. Extensive Margin: Trade-Link Survival Under Geopolitical Stress

4.4.1. Baseline survival effects

The intensive-margin results demonstrate that geopolitical risk reduces the value of existing trade flows within preferential agreements. A natural follow-up question is whether GPR also destroys trade relationships entirely – that is, whether it operates on the extensive margin by eliminating bilateral product-level trade links. Table 4 presents linear probability model (LPM) estimates of the survival probability for at-risk trade links (pair–HS2 combinations with positive trade in year t−1), using the same four-dimensional fixed-effect structure as the PPML specifications. The joint specification with lagged link-size controls (Column 5) reveals a striking decomposition. The GPR × DTA Depth interaction is −0.0052 (p = 0.007), indicating that each additional ten legally enforceable provisions reduce the one-year survival probability of an established intermediate-goods trade link by 0.52 percentage points when global geopolitical risk increases by one standard deviation. Meanwhile, the GPR × FTA interaction turns positive at +0.0028 (p = 0.086), suggesting that, conditional on agreement depth, the binary presence of an FTA provides a modest survival premium.

Evaluating the joint marginal effect reveals a critical institutional threshold. The net conditional response of link survival to a one-standard-deviation GPR shock is given by $\partial \Pr(\text{Survival})/\partial \text{GPR} = 0.28 - 0.52 \times (\text{Depth}/10)$ percentage points. This net derivative crosses zero at approximately 5.4 legally enforceable provisions. Below this threshold, the basic preferential tariff advantage dominates, conferring a modest survival premium. Above 5.4 provisions, the institutional rigidity and compliance penalty rapidly outweighs the tariff margin. Because the sample median depth among active preferential agreements is 12 provisions (Table

2), the vast majority of operational trade agreements lie well to the right of this turning point, placing them in the zone of net heightened vulnerability.

Table 4. Trade-Link Survival: GPR × Trade Agreement Interactions (LPM)

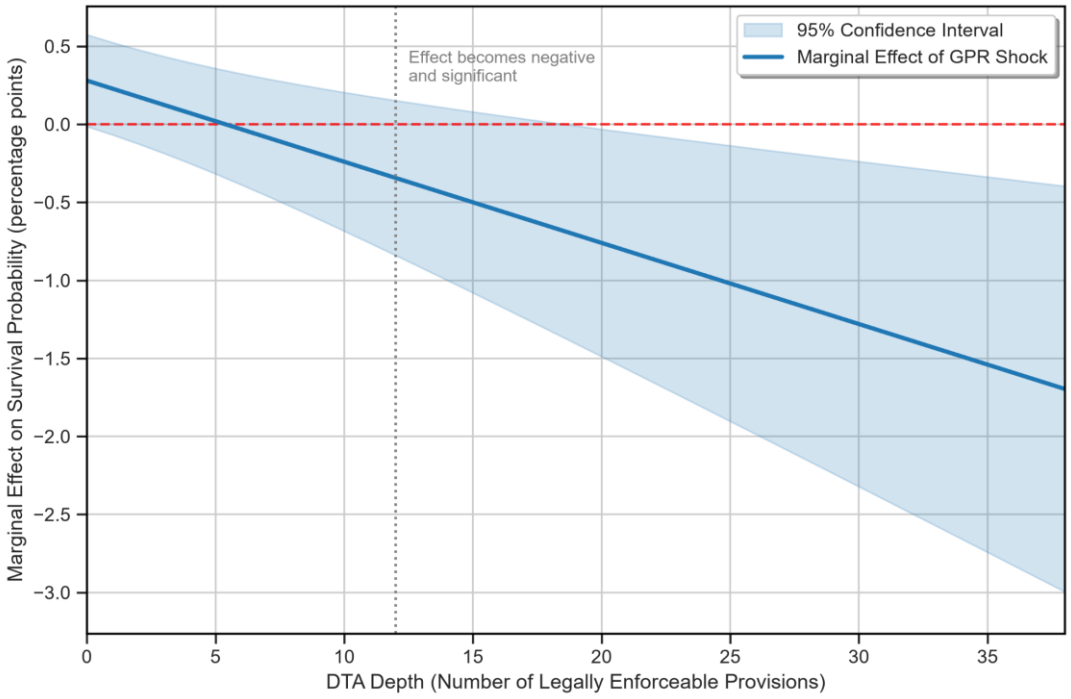
Variable	(1) FTA, no size	(2) FTA + size	(3) Depth, no size	(4) Depth + size	(5) Joint + size
GPR × DTA depth	–	–	−0.0040*	−0.0042*	−0.0052***
			(0.0021)	(0.0021)	(0.0017)
			[−0.40 pp]	[−0.42 pp]	[−0.52 pp]
GPR × FTA	−0.0044	−0.0047	–	–	0.0028*
	(0.0036)	(0.0037)			(0.0015)
	[−0.44 pp]	[−0.47 pp]			[+0.28 pp]
Lagged link-size control	No	Yes	No	Yes	Yes
Pair–HS2 fixed effects	Yes	Yes	Yes	Yes	Yes
Exporter–year fixed effects	Yes	Yes	Yes	Yes	Yes
Importer–year fixed effects	Yes	Yes	Yes	Yes	Yes
HS2–year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	8,045,131	8,045,131	8,045,131	8,045,131	8,045,131

Variable	(1) FTA, no size	(2) FTA + size	(3) Depth, no size	(4) Depth + size	(5) Joint + size

*Notes: LPM estimates. Standard errors in parentheses (pair-plus-year multiway clustered). Effects in percentage points. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Source: Authors' compilation based on CEPII BACI, World Bank DTA, and Caldara and Iacoviello (2022).

Figure 3. Marginal Effect of Geopolitical Risk on Trade-Link Survival Conditional on DTA Depth



Source: Authors' compilation based on CEPII BACI, World Bank DTA, and Caldara and Iacoviello (2022).

The mechanism underlying this extensive-margin vulnerability differs from the intensive-margin channel. While the trade-value margin reflects reduced trade volumes among continuing relationships, the extensive margin captures complete relationship termination. Figure 3 plots the marginal effect of GPR on link survival across the continuous range of DTA depth. The effect is mildly positive for shallow FTAs but slopes downward, becoming

significantly negative for deep agreements (those with more than 12 enforceable provisions). Deeper agreements impose higher fixed compliance costs – rules of origin documentation, regulatory conformity assessments, and standards certification – that become prohibitively expensive for smaller or more marginal trade links when geopolitical uncertainty raises the option value of waiting. The result is a selective pruning of the weakest trade links within deep-agreement networks, even as the core high-value relationships persist.

4.4.2. Robustness of survival estimates

We subject the survival findings to extensive robustness analysis (Table 5). Alternative constructions of global geopolitical risk demonstrate that the extensive-margin penalty is robust across three of the four shock variants (Table 5, Panel A). Specifically, negative and statistically significant depth interactions emerge under the threat component (−0.34 pp, p = 0.002), the annual peak shock (−0.65 pp, p = 0.007), and the log-transformed annual mean (−0.53 pp, p = 0.008). The sole exception is the realised acts component, which yields a negative but statistically insignificant coefficient (−0.37 pp, p = 0.229). Far from weakening our identification, this divergence provides compelling support for the real-options mechanism: firms defer irreversible investments and prune vulnerable supply relationships in response to anticipated risks (threats) rather than to past materialized events (acts).

Panel B of Table 5 demonstrates that the survival penalty is heavily concentrated among marginal, low-volume trading relationships. When the sample is restricted to established links with lagged trade exceeding USD 100,000, the estimated depth penalty declines in magnitude by roughly three-quarters to −0.13 percentage points (p = 0.011), and remains at −0.14 percentage points (p = 0.019) when restricted to links exceeding USD 1 million. Rather than indicating fragility, this quantitative gradient provides direct empirical confirmation of the fixed-compliance-cost mechanism: because administrative burdens, rules-of-origin documentation, and conformity certifications operate as quasi-fixed annual overheads, they impose a disproportionate burden on smaller trade flows. Under heightened geopolitical risk,

firms selectively prune these marginal links while preserving core, high-volume supply relationships.

Panel C of Table 5 evaluates the depth interaction exclusively within the subsample of country pairs that maintained an active preferential trade agreement in the preceding year. This intensive comparison isolates the marginal effect of agreement depth conditional on treaty existence, removing the non-preferential baseline group entirely. Under this restriction, the point estimate remains remarkably stable at −0.46 percentage points (compared to −0.52 percentage points in the full panel), though the substantial loss of identifying variation—a 57 percent contraction in sample size from 8.05 million to 3.48 million observations—inflates the multiway clustered standard error, rendering the coefficient marginally insignificant at conventional levels ($p = 0.162$). The economic consistency of the point estimate indicates that the survival penalty is driven by variation in institutional depth itself rather than by selection into baseline agreement status

Most compellingly, Figure 4 displays a leave-one-year-out Jackknife sensitivity analysis. Across all 16 iterations – each omitting a single calendar year from the sample – the GPR × Depth interaction remains strictly negative, ranging from −0.68 to −0.40 percentage points, and achieves $p < 0.01$ under pair clustering in every iteration. Under the more conservative pair-plus-year clustering, the coefficient is significant at $p < 0.10$ in all 16 iterations and at $p < 0.05$ in 15 of 16. This extraordinary temporal stability eliminates concerns that any single geopolitical event (e.g., the Russia–Ukraine conflict in 2022, the COVID-19 pandemic in 2020, or the Arab Spring in 2011) drives the aggregate finding.

Table 5. Survival Robustness: Alternative GPR Measures and Sample Restrictions

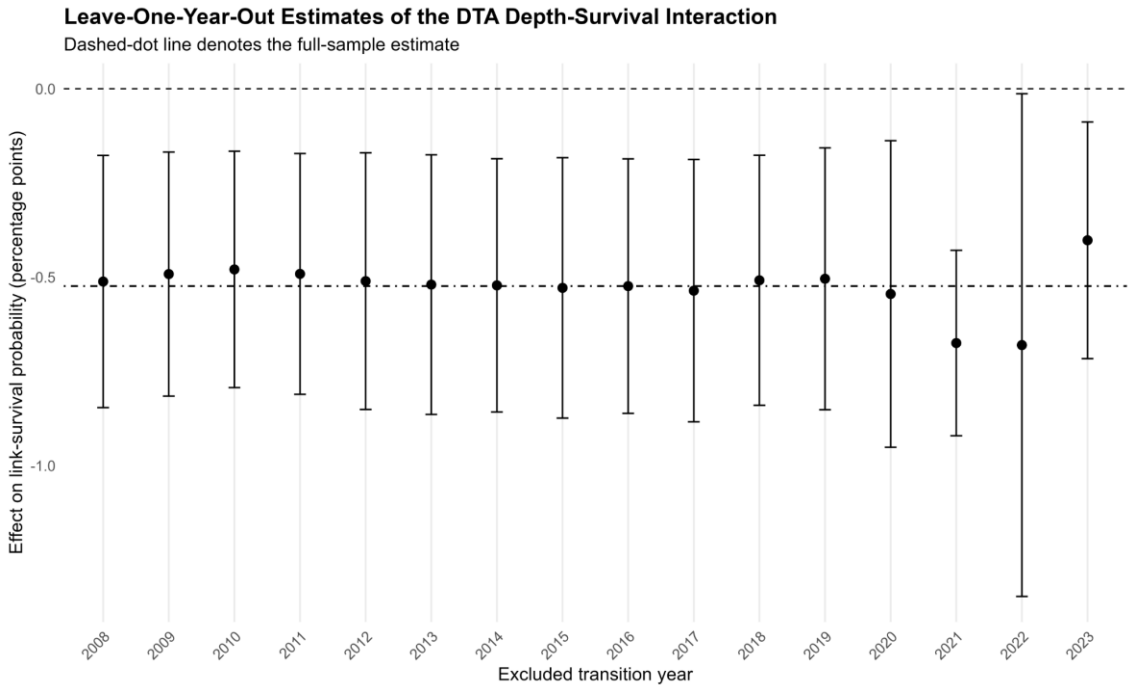
Panel	Check	Estimate	SE	Effect (pp)	p-value	N

A. Alternative global GPR	Baseline annual mean GPR	−0.0052	0.0017	−0.52	0.007	8,045,131
	Log annual mean GPR	−0.0053	0.0017	−0.53	0.008	8,045,131
	GPR threats	−0.0034	0.0009	−0.34	0.002	8,045,131
	GPR acts	−0.0037	0.0030	−0.37	0.229	8,045,131
	Annual peak GPR	−0.0065	0.0021	−0.65	0.007	8,045,131
B. Established-link samples	Lagged trade ≥ USD 100,000	−0.0013	0.0004	−0.13	0.011	3,842,229
	Lagged trade ≥ USD 1 million	−0.0014	0.0006	−0.14	0.019	2,079,423
C. Active-FTA-only sample	FTA-only, size-adjusted	−0.0046	0.0031	−0.46	0.162	3,479,262

*Notes: Standard errors are pair-plus-year multiway clustered. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

Source: Authors' compilation

Figure 4. Leave-One-Year-Out Estimates of the DTA Depth–Survival Interaction



Notes: Leave-one-year-out estimates of the GPR × DTA Depth interaction. Vertical bars denote 95% confidence intervals.

Source: Authors' compilation

4.5. Post-Exit Recovery Dynamics: Short-Term Friction Versus Medium-Term Resilience

Having documented that geopolitical risk accelerates trade-link destruction within deep agreements, we now examine the recovery dynamics following link exits. Table 6 reports LPM estimates for four post-exit outcomes: re-entry within one year, re-entry within two years, full value recovery within one year, and full value recovery within two years. The sample comprises 1,151,310 exit events observed between 2008 and 2022, with directed-pair fixed effects as the primary specification.

The results reveal a temporal asymmetry that provides further insight into the economic mechanism. For one-year re-entry, the GPR × Depth interaction is −0.0035 ($p = 0.044$, effect: −0.35 percentage points), indicating that deeper agreements slow the short-term reconstitution of disrupted trade links. However, for two-year re-entry, the coefficient collapses to statistical insignificance (+0.0001, $p = 0.978$), and the full-recovery outcomes at both horizons are similarly insignificant ($p > 0.40$).

This dynamic pattern is consistent with a fixed-cost adjustment mechanism. In the immediate aftermath of a geopolitical shock, the compliance and renegotiation costs embedded in deep agreements – re-establishing rules-of-origin documentation, renewing regulatory certifications, and rebuilding institutional partner linkages – create short-run friction that delays re-entry relative to shallow or non-preferential alternatives. At the two-year horizon, however, the depth interaction collapses to a precisely estimated zero (+0.0001, p = 0.978), and the full-value recovery outcomes are similarly indistinguishable from zero across both horizons (p > 0.40). These findings demonstrate an absence of permanent hysteresis: the institutional friction that delays reconstitution of severed links dissipates over the medium term. We emphasize that this represents the fading of an initial barrier rather than active institutional acceleration of re-entry – deep agreements create transitory adjustment frictions without erecting permanent structural impediments to relationship renewal.

Appendix Table A4 reports net GPR interaction effects at representative depth levels (0, 10, 20, and 30 enforceable provisions). For one-year re-entry, the net effect turns significantly negative only at high depth levels (20 provisions: −0.57 pp, p = 0.003; 30 provisions: −0.93 pp, p = 0.007), reinforcing the interpretation that the short-term friction is proportional to institutional complexity rather than a binary FTA effect.

Table 6. Post-Exit Recovery: GPR × DTA Depth Interactions (LPM)

Outcome	Directed-pair FE	N	Pair–HS2 FE	N
Re-entry within one year	−0.0035**	1,151,310	−0.0001	839,585
	(0.0016)		(0.0021)	
	[−0.35 pp]		[−0.01 pp]	
Re-entry within two years	0.0001	1,064,542	0.0013	752,846
	(0.0035)		(0.0056)	
	[+0.01 pp]		[+0.13 pp]	

Outcome	Directed-pair FE	N	Pair–HS2 FE	N
Full recovery within one year	−0.0009	1,151,310	0.0005	839,585
	(0.0011)		(0.0016)	
	[−0.09 pp]		[+0.05 pp]	
Full recovery within two years	−0.0002	1,064,542	0.0041	752,846
	(0.0036)		(0.0059)	
	[−0.02 pp]		[+0.41 pp]	

*Notes: LPM estimates. Standard errors in parentheses (pair + exit-year clustered). Effects in percentage points. *** p < 0.01, ** p < 0.05, * p < 0.10.*

Source: Authors' compilation

4.6. Country-Specific Geopolitical Risk

Panel A of Table 7 demonstrates that the intensive-margin buffering capacity of deep trade agreements emerges decisively once geopolitical risk is measured at the bilateral country-pair level. Using the average pair-country GPR, the baseline FTA interaction intensifies to −0.0852 (p = 0.007, effect: −8.17%), while the GPR × DTA Depth interaction turns positive and statistically significant at +0.0280 (p = 0.018, effect: +2.84%). The log-average and maximum pair-risk metrics yield virtually identical estimates (+2.81% and +2.43%, respectively).

A crucial econometric question is whether the emergence of depth significance reflects non-random sample selection, as the 44 economies with bilateral GPR series constitute systematically larger and more integrated markets (946 unordered pairs). The restricted-sample global GPR benchmark in Row 4 directly resolves this question. Restricting the sample while maintaining global GPR moves the depth coefficient from +0.0099 to +0.0160 (p = 0.143), accounting for 34 percent of the overall shift. Transitioning to bilateral country-specific risk accounts for the remaining 66 percent of the parameter movement (+0.0120) and the entire gain in statistical precision. The buffering capacity of institutional depth is therefore a substantive

economic consequence of geographic alignment between treaty disciplines and bilateral risk exposures, rather than an artifact of sample composition.

Crucially, contrasting Panel A and Panel B uncovers the central theoretical insight of this paper: a stark divergence between the intensive and extensive margins of international trade. Under identical bilateral risk metrics, institutional depth provides robust protection for the value of continuing intermediate-goods flows (+2.84% per ten provisions, $p = 0.018$), but systematically accelerates the destruction of bilateral product links (−0.17 pp per ten provisions, $p < 0.001$). Trade agreement depth protects relationship value while simultaneously pruning relationship count. Any empirical evaluation of preferential agreements that focuses exclusively on aggregate trade values will mistakenly conclude that deep institutions confer uniform resilience, completely overlooking the systemic thinning of the underlying supply network.

Table 7. Country-Specific GPR Robustness

Shock measure	GPR × FTA	GPR × DTA depth	N
Panel A. Intermediate-goods trade value (PPML)			
Average pair-country GPR	−0.0852***	0.0280**	2,520,781
	(0.0319)	(0.0118)	
	[−8.17%]	[+2.84%]	
Log-average pair-country GPR	−0.0843***	0.0277**	2,520,781
	(0.0294)	(0.0111)	
	[−8.09%]	[+2.81%]	
Maximum pair-country GPR	−0.0796**	0.0240*	2,520,781
	(0.0339)	(0.0129)	
	[−7.65%]	[+2.43%]	

Restricted-sample global GPR	−0.0617***	0.0160	2,520,781
	(0.0217)	(0.0109)	
	[−5.98%]	[+1.62%]	
Panel B. Trade-link survival (LPM)			
Average pair-country GPR	−0.0006	−0.0017***	1,878,443
	(0.0020)	(0.0004)	
	[−0.06 pp]	[−0.17 pp]	
Log-average pair-country GPR	−0.0014	−0.0015***	1,878,443
	(0.0019)	(0.0004)	
	[−0.14 pp]	[−0.15 pp]	
Maximum pair-country GPR	0.0007	−0.0020***	1,878,443
	(0.0021)	(0.0005)	
	[+0.07 pp]	[−0.20 pp]	

*Notes: Standard errors in parentheses (pair-plus-year multiway clustered). *** p < 0.01, ** p < 0.05, * p < 0.10.*

Source: Authors' compilation

4.7. Discussion: Mechanism Synthesis

The empirical findings, taken together, reveal a coherent and previously undocumented mechanism through which geopolitical risk interacts with deep trade agreements to reshape global value chain trade. We synthesize the evidence into three interconnected transmission channels.

4.7.1. The supply-chain interdependence channel

The first and most prominent mechanism operates through supply-chain interdependence. Preferential trade agreements – particularly those that eliminate tariffs and harmonize standards – encourage firms to organize production across borders in vertically fragmented value chains. This integration creates efficiency gains during normal times but

generates vulnerability during geopolitical crises. The sectoral evidence demonstrates that this channel is strongest in industries with high asset specificity, extensive cross-border component sourcing, and limited input substitutability – precisely the characteristics that international trade theory associates with hold-up problems under uncertainty (Antràs, 2003; Antràs & Helpman, 2004). When geopolitical events threaten the continuity of cross-border transactions, firms embedded in deep supply-chain relationships face adjustment costs that are proportional to the degree of prior integration.

4.7.2. The institutional quality channel

The second mechanism operates through specific institutional provisions that create shock-absorbing capacity. The provision-level analysis identifies competition policy, environmental standards, and research-and-technology cooperation as the disciplines that confer measurable resilience. These provisions share a common feature: they create durable institutional infrastructure – regulatory agencies, compliance monitoring systems, joint technology platforms – that continues to function during crises and reduces the transaction costs of maintaining trade relationships under heightened uncertainty. This finding is consistent with the incomplete-contracts theory of trade agreements (Horn, Maggi, & Staiger, 2010), which predicts that enforceable commitments in areas prone to opportunistic behavior are most valuable precisely when external shocks increase the temptation to renege on cooperative arrangements.

4.7.3. The fixed-cost adjustment channel

The third mechanism explains the extensive-margin dynamics documented earlier. Deep agreements embed substantial fixed costs of compliance – rules of origin, regulatory conformity assessments, and standards certification – that must be maintained to preserve preferential market access. Under normal conditions, these costs are justified by the margin of preference; however, when geopolitical risk rises, the expected present value of future trade declines, causing marginal trade links to fall below the viability threshold. The result is a selective

destruction of smaller trade links, which we observe as reduced survival probability. Finally, the post-exit recovery evidence indicates that this extensive-margin destruction does not induce permanent network hysteresis: the initial friction attenuates by the second year, yielding a zero net interaction (+0.0001, p = 0.978). We qualify this recovery dynamic by noting that the initial one-year friction is sensitive to the fixed-effects structure: while statistically significant under directed-pair fixed effects (−0.35 pp, p = 0.044), it attenuates to −0.01 pp (p = 0.951) under stringent pair–HS2 fixed effects. We therefore treat the post-exit recovery evidence with appropriate caution, resting the primary conclusions of the paper on the trade-link survival results, which remain remarkably robust across all baseline, sample, and shock specifications.

5. Conclusion

5.1. Policy implications

The findings carry implications for how trade agreements are evaluated, how their compliance requirements are designed, and how they figure in the resilience debate.

First, the evaluation of trade agreements must measure both margins. Ex-post assessments track trade value, and on that metric a deep agreement under geopolitical stress looks like insurance: value holds up. Beneath it the network is thinning, one small link at a time, and nothing in the value series records the loss. Monitoring frameworks, and any test of the premise that friend-shoring makes partner supply chains more resilient (Javorcik et al., 2024), should therefore report the number and survival of active pair–product links alongside value.

Second, the compliance cost of preferential status is the operative policy lever, and it should be scaled to relationship size. The links that disappear are the small and young ones, for which a fixed annual cost of origin certification, conformity assessment and documentation is largest per unit of trade (Anson et al., 2005; Hakobyan, 2015). Self-certification of origin, de minimis thresholds, simplified procedures for low-value consignments and broader cumulation reduce that cost without altering the preference margin, and the results here suggest their value rises precisely when uncertainty does.

Third, diversification is not a by-product of deep integration. Openness lowers volatility when shocks are country-specific and can be pooled across partners (Caselli et al., 2020). Global geopolitical risk is a common shock, and a bloc of deep agreements concentrates sourcing among partners exposed to it together, so a friend-shoring strategy built on institutional depth may raise the co-movement of supply-chain trade rather than reduce risk. Resilience should therefore be assessed as an objective separate from integration rather than assumed to accompany it, the more so because the sign of optimal resilience policy is itself model-dependent (Grossman et al., 2023).

Fourth, the results do not identify which provisions do the work. The provision-level screen singles out competition policy, environmental standards and research cooperation, but none of these coefficients survives correction for multiple testing. They are hypotheses for a focused design, not a template for negotiation.

Overall, the policy message is that institutional depth is not a substitute for resilience policy. Deep integration buys efficiency; it does not buy resilience, and on the evidence here the two can trade off against each other.

5.2. Limitations and prospects

Despite its contributions, this study is subject to several limitations, each of which points to a specific extension.

One limitation concerns the assignment of depth. Pairs with dense production-network trade sign deeper agreements (Orefice & Rocha, 2014), and pair–chapter fixed effects with lagged treatment absorb only the time-invariant part of that selection (Baier & Bergstrand, 2007). Measuring depth at the date of signature, or an event study around entry into force, would test whether the effect predates the agreement's own reshaping of the relationship.

Another concerns the shock. Global geopolitical risk has no cross-sectional variation, so identification rests entirely on interactions with exposure. The country-specific extension relaxes this, but on 44 economies and 946 pairs that are systematically larger than the full panel,

so its stronger intensive-margin estimates are read against a global-risk benchmark re-estimated inside the same subsample rather than against the full-panel results. A design that separates anticipated from realised conflict, which the threat and act components only approximate, would sharpen the shock itself.

The unit of observation is a pair–chapter link, not a firm: an exit may be one exporter leaving a market or every exporter leaving it. Transaction-level customs data would separate the firm margin from the product margin and allow compliance-cost heterogeneity by firm size, on which selective pruning turns, to be tested directly rather than inferred from link size.

Inference carries two econometric caveats. The fitted values of the linear probability model are not bounded to the unit interval, and a discrete-time hazard with the same fixed-effect structure is the natural check (Hess & Persson, 2012). Poisson estimation with four sets of fixed effects faces an incidental-parameter problem for which the available corrections are derived for three-way models (Weidner & Zylkin, 2021). The survival result does not depend on marginal significance; the intensive-margin estimates are less comfortably placed.

The data record the death of a link but not its replacement. An importer that loses a supplier in one partner may re-source the same chapter from a third country, so pair-level destruction may be importer-level diversion. Following importer–chapter imports across all origins after an exit would show whether pruning reduces total sourcing or merely relocates it.

Finally, the mechanism is inferred from a conjunction of signs across two margins, not observed. A direct test requires data on preference utilisation, so that links which stop claiming preferences can be distinguished from links which stop trading. That distinction is where the next study of deep agreements under geopolitical risk should begin.

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APPENDIX

Appendix Table A1. Sector Heterogeneity: GPR × Trade Agreement Interactions by HS2 Manufacturing Group

Sector	GPR × FTA	BH-FDR p	GPR × DTA depth	BH-FDR p	N
Textiles and footwear	−0.0447*** (0.0124) [−4.37%]	0.003	−0.0201*** (0.0058) [−1.99%]	0.005	2,522,056
Transport equipment	−0.0474** (0.0188) [−4.63%]	0.026	−0.0184* (0.0085) [−1.83%]	0.062	669,321
Precision and other manufactures	−0.0320** (0.0123) [−3.15%]	0.026	−0.0173*** (0.0053) [−1.72%]	0.006	1,171,921
Machinery and electronics	−0.0349** (0.0131) [−3.43%]	0.026	−0.0135* (0.0068) [−1.34%]	0.077	723,655
Plastics, rubber and leather	−0.0194** (0.0078) [−1.92%]	0.026	−0.0108* (0.0049) [−1.07%]	0.062	947,987
Chemicals	−0.0174 (0.0109) [−1.73%]	0.158	−0.0058 (0.0049) [−0.58%]	0.291	1,847,156
Stone, glass and base metals	−0.0235 (0.0163) [−2.32%]	0.187	−0.0022 (0.0069) [−0.22%]	0.749	2,427,994

Sector	GPR × FTA	BH-FDR p	GPR × DTA depth	BH-FDR p	N
Agro-food inputs	0.0202 (0.0124) [+2.04%]	0.158	−0.0039 (0.0061) [−0.39%]	0.579	3,184,818
Minerals and fuels	0.0018 (0.0189) [+0.18%]	0.923	0.0006 (0.0097) [+0.06%]	0.949	497,298
Wood, pulp and paper	0.0296 (0.0188) [+3.00%]	0.158	0.0116 (0.0088) [+1.17%]	0.268	2,760,150

*Notes: Standard errors two-way clustered by pair and year in parentheses. Asterisks denote BH-FDR q-values: *** q < 0.01, ** q < 0.05, * q < 0.10.*

Source: Authors' compilation

Appendix Table A2. DTA Mechanism Bundles: GPR × Bundle Depth Interactions (FTA-Conditional)

Bundle	Estimate	SE	Full-bundle effect (%)	Raw p	BH-FDR p	N
Environmental provisions	0.0384	0.0195	+3.91%	0.049	0.244	16,752,356
Digital and knowledge	0.0453	0.0338	+4.63%	0.180	0.451	16,752,356
Border coordination	0.0178	0.0213	+1.80%	0.404	0.673	16,752,356
Investment and production	0.0072	0.0180	+0.72%	0.691	0.753	16,752,356
Services and procurement	0.0059	0.0188	+0.59%	0.753	0.753	16,752,356

Notes: Standard errors two-way clustered by pair and year in parentheses. Asterisks denote BH-FDR q-values. The environmental bundle contains only environmental laws.

Source: Authors' compilation

Appendix Table A3. Individual DTA Provision Screening: GPR × Provision Interactions (FTA-Conditional)

Provision	Estimate	SE	Effect (%)	Raw p	BH-FDR p	N
Competition policy	0.0353	0.0169	+3.60%	0.037	0.228	16,752,356
Environmental laws	0.0384	0.0195	+3.91%	0.049	0.228	16,752,356
Research and technology	0.0343	0.0168	+3.49%	0.041	0.228	16,752,356
Public procurement	0.0330	0.0208	+3.36%	0.112	0.313	16,752,356
Movement of capital	0.0215	0.0123	+2.17%	0.081	0.283	16,752,356
Technical barriers to trade	0.0205	0.0173	+2.08%	0.235	0.412	16,752,356
Intellectual property rights	0.0160	0.0188	+1.61%	0.395	0.503	16,752,356
Trade-related intellectual property	0.0160	0.0141	+1.61%	0.255	0.412	16,752,356
SPS measures	0.0150	0.0162	+1.51%	0.355	0.497	16,752,356
Data protection	0.0146	0.0131	+1.47%	0.265	0.412	16,752,356
Trade-related investment measures	0.0004	0.0100	+0.04%	0.969	0.969	16,752,356
Customs	−0.0024	0.0154	−0.24%	0.877	0.945	16,752,356
Investment	−0.0119	0.0152	−1.18%	0.437	0.509	16,752,356
Trade in services	−0.0207	0.0168	−2.05%	0.216	0.412	16,752,356

Notes: Standard errors two-way clustered by pair and year in parentheses. Asterisks denote BH-FDR q-values. Excludes information society and innovation policies (< 500 pairs)

Source: Authors' compilation

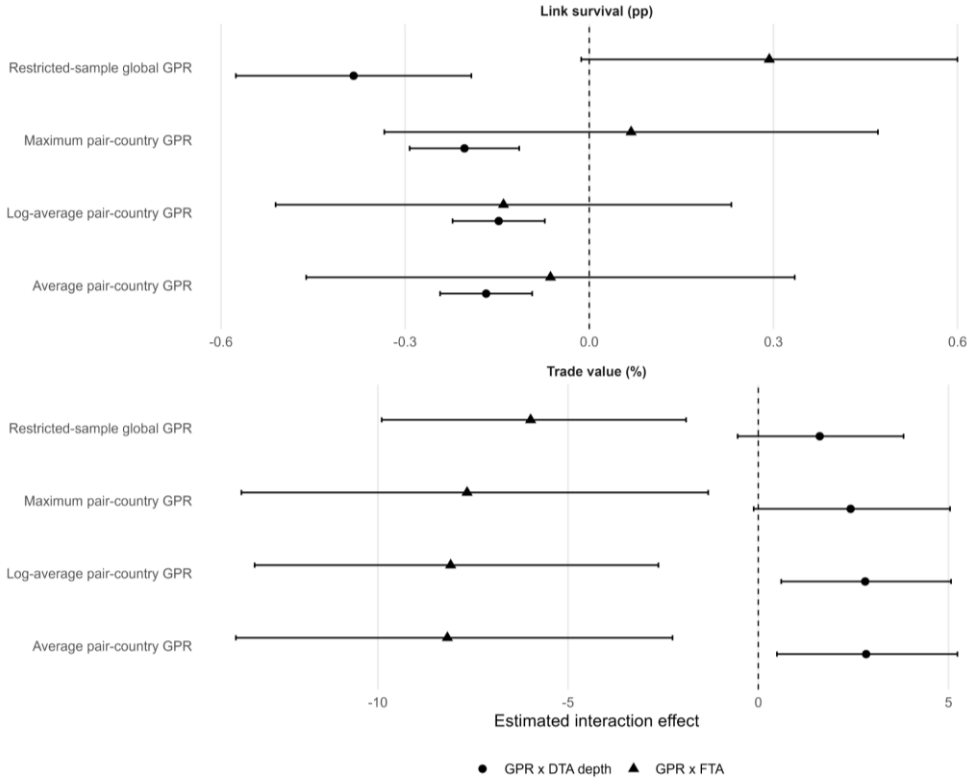
Appendix Table A4. Post-Exit Recovery: Net GPR Interaction Effects at Representative Depth Levels

Outcome	Enforceable provisions	Net effect (pp)	p-value	N
Re-entry within one year	0	+0.14 pp	0.570	1,151,310
	10	−0.22 pp	0.089	1,151,310
	20	−0.57 pp	0.003	1,151,310
	30	−0.93 pp	0.007	1,151,310
Re-entry within two years	0	+0.35 pp	0.567	1,064,542
	10	+0.36 pp	0.451	1,064,542
	20	+0.37 pp	0.527	1,064,542
	30	+0.38 pp	0.653	1,064,542
Full recovery within one year	0	−0.02 pp	0.912	1,151,310
	10	−0.11 pp	0.291	1,151,310
	20	−0.19 pp	0.116	1,151,310
	30	−0.28 pp	0.187	1,151,310
Full recovery within two years	0	−0.07 pp	0.901	1,064,542
	10	−0.10 pp	0.826	1,064,542
	20	−0.12 pp	0.827	1,064,542
	30	−0.14 pp	0.863	1,064,542

*Notes: Standard errors two-way clustered by pair and year in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10*

Source: Authors' compilation

Appendix Figure A1. Country-Specific GPR Robustness: Trade Value and Link Survival



Notes: Point estimates and 95% confidence intervals for country-specific GPR specifications

Source: Authors' compilation